# Do Transcriptomic Foundation Models Beat PCA?

## Benchmarking BulkFormer for Blood-Based Classification of Depression

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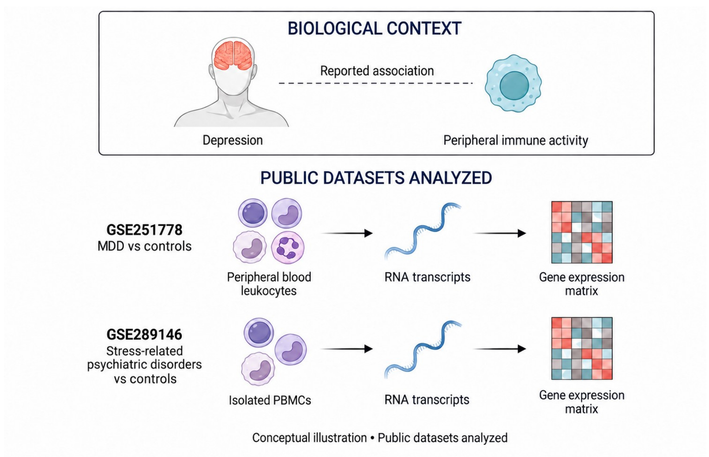
Grand Valley State University
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## Background & Research Question

Major depressive disorder (MDD) is diagnosed clinically, with no established molecular diagnostic test. Foundation models pretrained on large gene-expression datasets may learn transferable features, but whether they outperform simple baselines in small clinical cohorts remains unclear. We compared BulkFormer-37M (Kang et al., 2026) with principal component analysis (PCA), which summarizes variation across thousands of genes.

**Does BulkFormer improve classification beyond PCA?**

## Two Public Cohorts




**Primary** — GSE251778 (Mokhtari et al., 2025): peripheral blood leukocytes from 80 adults with MDD and 89 controls. Sequencing protocols differed by sex.

**Exploratory** — GSE289146 (Rehawi et al., 2025): PBMCs from 229 participants classified as affected by stress-related psychiatric disorders and 70 controls.

## Representation Comparison

Same participants and measured genes, with ln(TPM + 1) expression.

**PCA**
Fitted within training folds
Retains 90% of training variance
Median PCs: 55 MDD / 169 mixed

**BulkFormer-37M**
Pretrained, not fine-tuned
Mean-pooled sample embedding
128 learned + 3 summaries

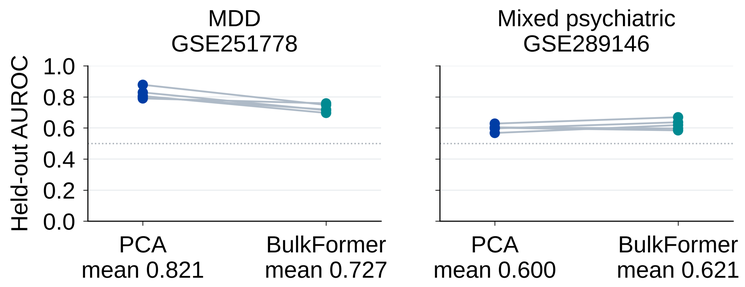
**Same ridge logistic regression with balanced class weights**

## Matched Evaluation

Both methods used the same participants and matched cross-validation folds. Inner folds selected the ridge penalty. Five outer folds generated held-out predictions, repeated across five partitions. We report mean pooled out-of-fold AUROC across repeats.

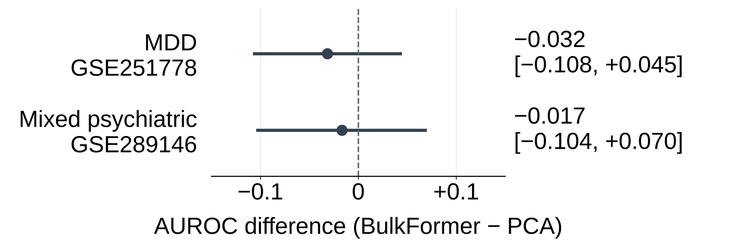
**AUROC**: 0.5 = chance ranking; 1.0 = perfect ranking.

## Main Result: No Consistent BulkFormer Gain



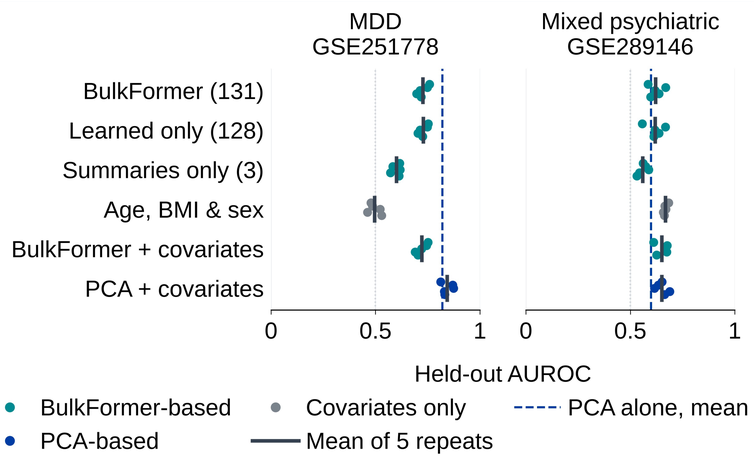

PCA had higher pooled AUROC in every MDD repeat. Mixed-cohort rankings varied across repeats: BulkFormer had higher mean pooled AUROC, while PCA had higher mean fold AUROC.

Each line is one repeat; repeats share participants (not independent).




Original partition (repeat 1; smallest MDD gap): paired 95% bootstrap intervals for BulkFormer − PCA included zero in both cohorts. Intervals use fixed held-out predictions and exclude retraining uncertainty.

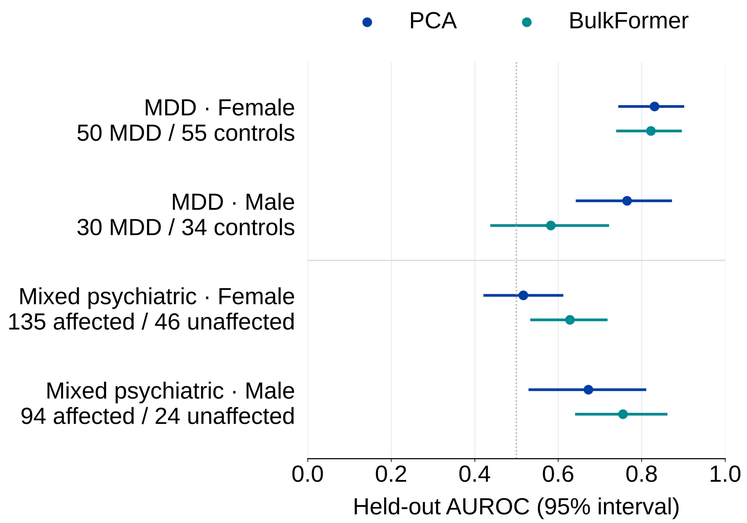
## Feature & Covariate Sensitivity




Learned-only features performed similarly to the full output; summaries alone were weaker. Age, BMI and sex increased PCA's mean pooled AUROC in MDD (0.821 to 0.843; four of five repeats); BulkFormer changed little (0.727 to 0.722). In the mixed cohort, covariates alone had higher mean pooled AUROC (0.668) than either combined model (≈0.651).

Points show five CV repeats; their spread is not a confidence interval.

## Exploratory Models by Recorded Sex




Separate-sex models ranked differently across subgroups, using one partition each; intervals exclude retraining uncertainty. Differences do not establish biological sex effects, and in MDD sex coincides with sequencing protocol.

## Interpretation & Biological Context

**BulkFormer-37M did not consistently outperform PCA in these two cohorts.**

PCA outperformed BulkFormer in MDD; in the mixed cohort, covariates alone exceeded both on average. Depression has been linked to low-grade inflammation (Osimo et al., 2019), but blood expression mixes cell composition and gene activity, which these analyses cannot separate.

## Limitations & Next Steps

- One BulkFormer checkpoint, linear classifier and no fine-tuning
- Cohorts differ in phenotype, cell type and protocol; mixed-cohort groups differ in age
- MDD sex/protocol confounding and fallback gene lengths in TPM reconstruction
- No external validation; pretraining overlap remains unverified

**Next**: test fine-tuning, cell-composition baselines and independent cohorts.

## Code, Reproducibility & Contact

Scan for code, predictions, figures and references.

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**References:** Kang et al. 2026; Mokhtari et al. 2025; Rehawi et al. 2025; Osimo et al. 2019 (full citations via QR). Illustration: BioRender.com.